independent existence and exhibit division of labour. Among the metazoans, cells may be functionally isolated or similar kinds of cells may be grouped together to form tissues, organ and organ systems.

Cellular level of organisation

This basic level of organisation is seen in sponges. The cells in the sponges are arranged as loose aggregates and do not form tissues, i.e. they exhibit cellular level of organisation. There is division of labour among the cells and different types of cells are functionally isolated. In sponges, the outer layer is formed of pinacocytes (plate-like cells that maintain the size and structure of the sponge) and the inner layer is formed of choanocytes. These are flagellated collar cells that create and maintain water flow through the sponge thus facilitating respiratory and digestive functions.

Animals such as sponges lack nervous tissue and muscle tissue, what does this tell you about sponges?

Tissue level of organisation

In some animals, cells that perform similar functions are aggregated to form tissues. The cells of a tissue integrate in a highly coordinated fashion to perform a common function, due to the presence of nerve cells and sensory cells. This tissue level of organisation is exhibited in diploblastic animals like cnidarians. The formation of tissues is the first step towards evolution of body plan in animals (*Hydra* - Coelenterata).

Organ level of organisation

Different kinds of tissues aggregate to form an organ to perform a specific function. Organ level of organisation is a further advancement over the tissue level of organisation and appears for the first time in the Phylum Platyhelminthes and seen in other higher phyla.

Organ system level of organisation

The most efficient and highest level of organisation among the animals is exhibited by flatworms, nematodes, annelids, arthropods, molluscs, echinoderms and chordates. The evolution of mesoderm in these animals has led to their structural complexity. The tissues are organised to form organs and organ systems. Each system is associated with a specific function and show organ system level of organisation. Highly specialized nerve and sensory cells coordinate and integrate the functions of the organ systems, which can be very primitive and simple or complex depending on the individual animal. For example, the digestive system of Platyhelminthes has only a single opening to the exterior which serves as both mouth and anus, and hence called an **incomplete digestive system**. From Aschelminthes to Chordates, all animals have a **complete digestive system** with two openings, the mouth and the anus.

Similarly, the circulatory system is of two types, the **open type**: in which the blood remains filled in tissue spaces due to the absence of blood capillaries. (arthropods, molluscs, echinoderms and urochordates) and the **closed type:** in which the blood is circulated through blood vessels of varying diameters (arteries, veins and capillaries) as in annelids, cephalochordates and vertebrates.

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